

Lecture #2: Linear Functions, Quadratic Functions, Compositions

MATH 145 – Calculus I

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Types of Functions

There are many different types or categories of functions. Recognizing their properties can be very important, as we can utilize their shared properties.

We will talk about three kinds today:

- Linear Functions
- Quadratic Functions
- Compositions

Linear Functions

Linear functions are the easiest type of function to understand. You've likely already seen a lot of them.

A function f is **linear** provided that its average rate of change is constant on every choice of interval in its domain. That is, for any inputs a and b for which $a \neq b$, it follows that

$$\frac{f(b) - f(a)}{b - a} = m,$$

for some fixed constant m .

We call m the **slope** of the linear function f .

Linear Functions

The formula for the function is $f(x) = mx + b$, where m and b are constants. (Do you recognize the meaning of m and b ?)

Example:

1. Find the formula for a line with slope $m = 3$ and passing through the point $(-2, 5)$.
2. Find the equation of the line passing through the two points $(-2, 1)$ and $(4, -3)$.

Formulas for Linear Functions

A line with slope m (or equivalently, average rate of change m) that passes through the point (x_0, y_0) has an equation of “**point-slope form**”:

$$y = y_0 + m(x - x_0).$$

For the line with slope m and passing through $(0, y_0)$, its equation in “**slope-intercept form**” is

$$y = y_0 + mx.$$

$(0, y_0)$, the point at which the line crosses the y -axis, is called the **y -intercept**, denoted as b .

Linear Function Example

1. A town's population initially has 28750 people present and then grows at a constant rate of 825 people per year. Find a linear model $P = f(t)$ for the number of people in the town in year t .
2. The amount of money a show on a broadcast network makes when they they sell the rights to Netflix is directly rated by a linear function $C(e)$ to how many episodes have been produced at the time of sale. A show with 39 episodes sold their rights for \$50 million, whereas a show with 106 episodes sold their rights for \$145 million. If a show with 58 episodes is put up for sale, how much will Netflix pay?

Quadratic Functions

A **quadratic function** is a function of the form $f(x) = ax^2 + bx + c$, where a , b and c are constants.

The graph of a quadratic function is called a **parabola**. Note that the rate of change in a quadratic function varies depending on the interval.

a determines whether it is **concave up** or **concave down**.

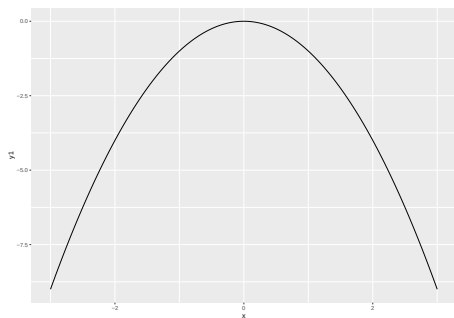


Figure 1: $a < 0$

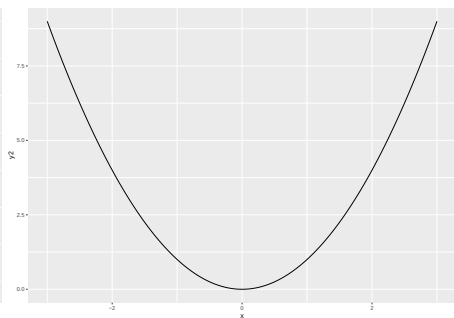


Figure 2: $a > 0$

Quadratic Formula

Let a , b and c be real numbers such that $a \neq 0$. The equation $ax^2 + bx + c = 0$ can have 0, 1 or 2 real solutions. The real solutions are given by the quadratic formula,

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a},$$

provided that $b^2 - 4ac \geq 0$.

1. If $b^2 - 4ac > 0$, $q(x)$ has two real solutions.
2. If $b^2 - 4ac = 0$, $q(x)$ has one real solutions.
3. If $b^2 - 4ac < 0$, $q(x)$ has no real solutions, the two roots of $q(x)$ are complex numbers (numbers of the form $z = a + bi$).

Graphs of Quadratic Functions

Example: If $f(x) = x^2 + 3x + 2$, solve $f(x) = 0$. Use this to sketch a graph of $f(x)$.

Notes:

1. $f(x_0) = 0$ if and only if f crosses the x -axis at x_0 .
2. You need $f(x) = 0$ to use the quadratic formula.

$$x^2 = 2x + 8 \Rightarrow x^2 - 2x - 8 = 0$$

3. Factorization also works, if the quadratic function can be factored into two expressions of degree 1.

Graphs of Quadratic Functions

The quadratic function $y = q(x) = ax^2 + bx + c$ has its vertex at the point

$$\left(-\frac{b}{2a}, q\left(-\frac{b}{2a}\right)\right).$$

When $a > 0$, the vertex is the lowest point on the graph of q , while if $a < 0$, the vertex is the highest point on the graph of q . Moreover, the graph of q is symmetric about the vertical line $x = -\frac{b}{2a}$.

Example: If $f(x) = x^2 + 3x + 2$, find the vertex.

Graphs of Quadratic Functions

A quadratic function with vertex (h, k) may be written in the form $y = a(x - h)^2 + k$. The constant a may be determined from one other function value for an input $x \neq h$.

Example: A quadratic function q has vertex $(2, 4)$, and passes through the point $(1, 3)$. Find the equation of $q(x)$, and find the roots (where $q(x) = 0$ of q).

A Quadratic Model

Height of an object falling under the force of gravity}

For an object tossed vertically from an initial height of s_0 feet with a velocity of v_0 feet per second, the object's height at time t (in seconds) is given by the formula

$$h(t) = -16t^2 + v_0t + s_0$$

For an object tossed vertically from an initial height of s_0 meters with a velocity of v_0 meters per second, the object's height at time t (in seconds) is given by the formula

$$h(t) = -4.9t^2 + v_0t + s_0$$

Note that $g = 9.8m/s = 32ft/s$.

Quadratic Function Example

Example: A baseball is tossed upward with an initial velocity of $2^m/s$ from a height of 3 meters. Find a formula $h(t)$ for the height of the ball at time t . When does it reach its highest point? When does it hit the ground?

Example: New York Yankee Giancarlo Stanton once hit a home run that came off the bat at 121.7 mph, the hardest hit home run ever recorded. (They started tracking this with Statcast in 2015.) How long would it have taken for the ball to hit the ground if he hit in in a flat field? Note that 1 mph is equivalent to $\frac{22}{15}$ feet per second.

Composition of Functions

Functions can get much more complicated than this.

If f and g are functions such that $g : A \rightarrow B$ and $f : B \rightarrow C$, we define the **composition of f and g** to be the new function $h : A \rightarrow C$ give by

$$h(t) = f(g(t)).$$

We also sometimes use the notation $h = f \circ g$, where $f \circ g$ is the single function defined by $(f \circ g)(t) = f(g(t))$.

Quadratic Function Example

Example: Let $f(x) = 2x + 3$, $g(x) = x^2 - 1$ and $h(x) = 2x^2$. Find:

1. $(g \circ f)(x)$.
2. $f(g(x))$.
3. $h(g(x))$.